SmartTea-LK · Component 04

AI-Based Tea Plantation Resource Optimization and Intervention Planning

Research report: problem, system design, hybrid dataset, methodology, preliminary results and work plan

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Project	SmartTea-LK research initiative (Sri Lankan tea sector), CSP 001
Date	8 October 2026
Status	Design and dataset complete; baseline models validated. Model tuning, optimizer, scheduler and application are in progress.

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Abstract

Sri Lankan tea estates still apply fertilizer, fungicide, irrigation and lime on fixed, rule-of-thumb calendars. These calendars ignore the weather that follows each application: a monsoon downpour can wash fertilizer off a steep slope within hours, and a dry spell can make an input ineffective. This component of the SmartTea-LK initiative designs an AI decision-support system that takes the *live* condition of each tea block and recommends **what to do next, how much, and when**. The goal is the highest growth from the smallest physical input, timed so the weather does not undo it.

The system is a three-tier pipeline. **Tier 1** predicts: XGBoost surrogate models estimate the yield response to a candidate action and the share of fertilizer that will be lost to runoff, using a live Open-Meteo forecast. **Tier 2** optimizes: the NSGA-II multi-objective algorithm searches for the amounts that maximize predicted yield while minimizing water and fertilizer, inside agronomic hard constraints. **Tier 3** schedules: an Action-Window scanner picks the best execution time in the next 72–96 hours. The scope is purely agronomic; cost, labour and machinery are deliberately excluded.

No public dataset combines block-level soil sensing, fertilizer timing and yield for Sri Lanka, so a **hybrid dataset** was built: real weather (ERA5), real rain timing, real issued forecasts, real soil (SoilGrids) and real national production figures form the environment, while management decisions and crop response are simulated with a documented process model. It covers 5 real tea-growing locations × 10 blocks over 2021–2025 (9,053 plucking rounds, 3,233 field interventions).

Baseline results are promising but honest. The XGBoost yield model reaches R^2 0.94 on a held-out year but only **R^2 0.66 on blocks it has never seen**. The runoff model reaches R^2 0.94 with hindsight knowledge of rainfall, but **R^2 0.62 when it has only the forecast**, which is how it must run live. This gap, caused by real forecast error (which under-predicts rain by about 4 mm/day), is the central research finding so far: forecast quality, not model capacity, limits how well inputs can be timed.

1. Introduction

1.1 Background

Tea is one of Sri Lanka's most important export crops. It is grown in three elevation bands with very different climates: high-grown (above about 1,200 m, e.g. Nuwara Eliya and Hatton), medium-grown (about 600–1,200 m, e.g. Kandy) and low-grown (below about 600 m, e.g. Ratnapura and Deniyaya). National output was 299.5 million kg in 2021 and fell to 249.9 million kg in 2022, when the chemical-fertilizer import ban and the economic crisis hit the sector. That drop is a reminder of how directly tea yield depends on nutrient management.

Most mature Sri Lankan tea is rain-fed and grown on steep slopes in high-rainfall areas. Under these conditions the *timing* of an input matters almost as much as its amount: fertilizer spread just before heavy rain is partly washed away, fungicide sprayed before a shower is ineffective, and irrigation on a day before rain is wasted.

1.2 Problem statement

Estates typically follow static schedules: fertilizer in fixed months, sprays when disease is visible, lime on a multi-year cycle. These schedules ignore the local microclimate and the forecast. The result is wasted inputs, nutrient runoff into streams, and lower yield than the same inputs could give if they were timed well.

1.3 Research question and objectives

Research question. How can the minimum physical resources (fertilizer, water, fungicide, lime) be applied to achieve maximum tea growth, and when should each be deployed so that the weather does not negate its effect?

Objectives:

- **O1.** Build surrogate models that predict (a) the yield response of a tea block to a candidate intervention and (b) the fraction of applied fertilizer lost to runoff, under *forecast* weather.
- **O2.** Formulate intervention planning as a multi-objective problem (maximize yield, minimize water and fertilizer) and solve it with NSGA-II under agronomic constraints.
- **O3.** Design an Action-Window scheduler that selects the best execution time in a 72–96 h forecast horizon.
- **O4.** Deliver the pipeline as a live system that re-plans when a new soil reading arrives and every hour as the forecast changes.
- **O5.** Evaluate the system against the fixed-calendar practice on resource use, runoff avoided and yield, and quantify the impact of forecast error.

1.4 Scope

- **In scope:** fertilizer, irrigation, fungicide and pH correction (dolomite); soil and weather conditions; yield response; timing.
- **Out of scope (deliberately):** cost, labour, machinery and budget variables. These were removed from the original framework so that the work measures pure agronomic efficiency.
- **Sensors:** no IoT hardware is built in this component. Soil readings are entered manually through an endpoint that accepts exactly the JSON a future IoT device will send, so real sensors can be added later without changing the system.

2. Related work and data availability

A search for public data found **no dataset that combines block-level tea soil sensing, fertilizer timing and yield for Sri Lanka**. The closest sources, and how this project uses them, are summarised below.

Table 1. Data sources and related studies.

Source	What it offers	Use in this project
Open-Meteo Historical API (ERA5 / ERA5-Land)	Hourly reanalysis weather and soil moisture at 3 depths, any location, 2021–2025	Real environment for all 5 estates; real rain timing after each action
Open-Meteo Previous Runs API	Archived forecasts at 1–4 day lead time (from Jan 2024)	Real forecast error; training the runoff model on forecast rain
NASA POWER	Coarse (0.5°) daily weather and soil wetness	Independent cross-check only
ISRIC SoilGrids 2.0 (250 m)	pH, total N, organic carbon, texture, CEC, bulk density	Real starting soil for each of the 50 blocks
Forbes & Walker monthly reports	National tea production by elevation, monthly	Realism check of simulated yield seasonality
Uva Medium rainfall–yield study (2003–2012)	Optimum 2,500–3,000 mm/yr; at least 50 mm per month	Calibration reference
Explainable AI for Ceylon tea yield (2025); NTHRI hybrid tea-yield study, Pakistan (2022)	ML yield modelling for tea	Related work; data not public
Kaggle “Crop Recommendation”	Crop-type classification	Not suitable: no tea, no timing

The best route to final validation is to obtain one estate's real fertilizer and yield records from the Tea Research Institute (TRI) or a plantation company. Until then, the hybrid dataset described in Section 4 is used.

3. System design

3.1 Three-tier AI pipeline

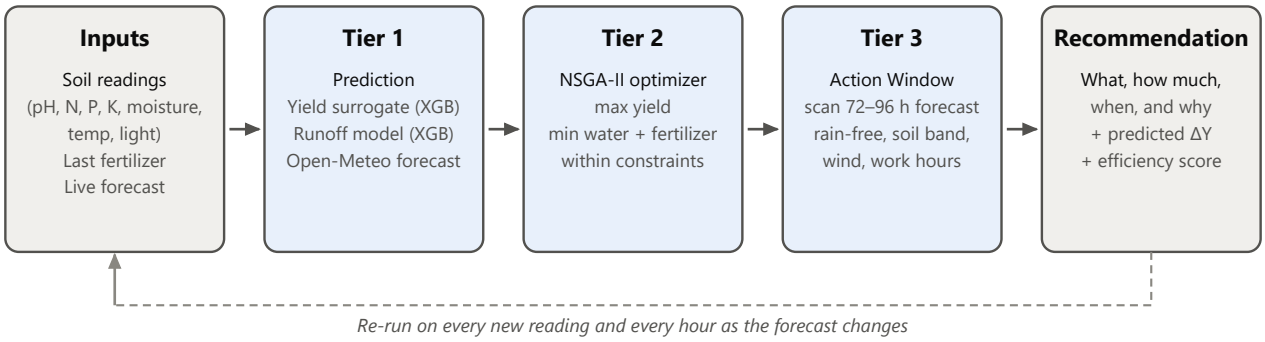


Figure 1. The three-tier pipeline. Each tier narrows the decision: Tier 1 predicts consequences, Tier 2 chooses amounts, Tier 3 chooses the time.

3.2 Inputs

Each block is described by static features (elevation, slope, soil texture, field capacity and wilting point, cultivar, bush age, shade, pruning cycle, irrigation availability) and by live readings. The live reading carries the same fields a future IoT soil node will send. The schema below is the **proposed** contract for `POST /api/readings`; it must stay stable once fixed.

```
{ "block_id": "E03-B07",
  "timestamp": "2026-10-08T06:00:00+05:30",
  "soil_moisture_pct": 34.2, "soil_temp_c": 19.8,
  "soil_ph": 5.1, "soil_n_ppm": 21.4, "soil_p_ppm": 18.0, "soil_k_ppm": 92.0,
  "light_lux": 41200, "sunshine_hours": 6.5,
  "last_fertilizer": { "product_code": "NPK_HN", "amount_kg_per_ha": 240, "date": "2026-07-14" } }
```

Live weather comes from the Open-Meteo forecast API (free, no key): hourly rain, rain probability, temperature, humidity, wind and radiation for each estate's coordinates.

3.3 Tier 1: prediction

- **Yield surrogate (XGBoost).** Predicts green-leaf yield per hectare per day for a plucking round from soil, weather, recent inputs and block features. It is called thousands of times by the optimizer, so it must be fast; XGBoost also handles missing values and can be explained with SHAP.
- **Runoff / efficiency model (XGBoost).** Predicts the percentage of applied fertilizer lost, from forecast rain (12/24/48/72 h), slope, soil moisture relative to field capacity, temperature and product. For live use it is trained on *forecast* rain columns, never on the rain that actually fell.
- **Weather.** The Open-Meteo forecast is used directly. An optional local correction model (XGBoost or LSTM trained on the forecast-vs-actual archive) may be added to remove the known dry bias.
- **Baselines.** Ridge regression, Random Forest, LightGBM and an MLP are trained for comparison.

3.4 Tier 2: NSGA-II multi-objective optimizer

For each block, NSGA-II (pymoo) searches over the candidate intervention amounts. The proposed formulation:

Table 2. Proposed NSGA-II formulation.

Element	Definition
Decision variables	Fertilizer amount (0, or 100–400 kg/ha mixture); irrigation depth (0–25 mm, irrigable blocks only); fungicide spray (yes/no); dolomite (yes/no, only when pH is below target)
Objective 1 (maximize)	Predicted yield change ΔY from the yield surrogate, with fertilizer N reduced by the predicted runoff loss
Objective 2 (minimize)	Total physical input: water + fertilizer (normalised so the two are comparable)
Hard constraints	From `07_operational_constraints.csv`: dose limits, annual N ceiling (300–330 kg N/ha/yr), ≥45 days between fertilizer, 30 days after pruning, reservoir capacity, pH target 4.5–5.5, 7-day pre-harvest interval after spraying
Search bounds	Kept inside the range seen in training data, because tree models cannot extrapolate
Choosing one plan	NSGA-II returns a Pareto front. A selection rule is still to be fixed (candidates: knee point; highest yield per kg of input; highest yield within a resource budget)

3.5 Tier 3: Action-Window scheduler

The scheduler scans each hour of the 72–96 h forecast, scores it with the runoff model on forecast rain, and rejects hours that break a rule. Soil-moisture rules are expressed relative to each block's own field capacity (FC) and wilting point (WP), because the real clay soils have a much higher wilting point than textbook defaults.

Table 3. Action-Window rules (from the operational constraints table).

Rule	Applies to	Value
Rain-free period after application	Fertilizer	≥ 12 h
Max forecast rain in next 12 h	Fertilizer	5 mm
Max forecast rain probability	Fertilizer	30 %
Soil-moisture band	Fertilizer	WP + 0.25·(FC − WP) to WP + 0.85·(FC − WP)
Rain-free period after spraying	Fungicide	≥ 4 h
Max wind speed (drift)	Fungicide	4 m/s
Irrigation trigger	Irrigation	below WP + 0.35·(FC − WP); max 25 mm per event
Working hours	All field work	start between 06:00 and 16:00
Forecast horizon	Scheduler	96 h

Open design question: where is timing decided? Tier 2 chooses amounts and Tier 3 chooses time, but yield itself depends on timing. Two options are being considered: (a) include the execution time as an NSGA-II decision variable, or (b) let Tier 3 re-score the predicted yield for each feasible window before the final choice.

3.6 Output

Each run returns one recommendation per block. The action is one of: apply or delay fertilizer, irrigate or skip, apply dolomite, spray fungicide, or no action.

```
{ "block_id": "E03-B07", "action": "Apply fertilizer", "amount_kg_per_ha": 240,
  "window": "2026-10-08 06:00–10:00",
  "reason": "Soil moisture in band; no rain next 14 h",
  "predicted_yield_change_pct": 9.2, "efficiency_score": 0.93 }
```

The original framework example (“15.5 kg, max 20 kg”) is unrealistic per hectare; real tea fertilizer is about 150–350 kg/ha of mixture per application, 3–4 times a year. The examples in this report use realistic rates.

3.7 Software architecture

The SmartTea-LK project fixes the application stack: **React.js with Tailwind CSS** (frontend), **Java + Spring Boot** (backend) and **MongoDB** (database). The AI tiers run in a separate **Python ML service** (FastAPI), because Java has no practical equivalent of XGBoost training, pymoo or SHAP.

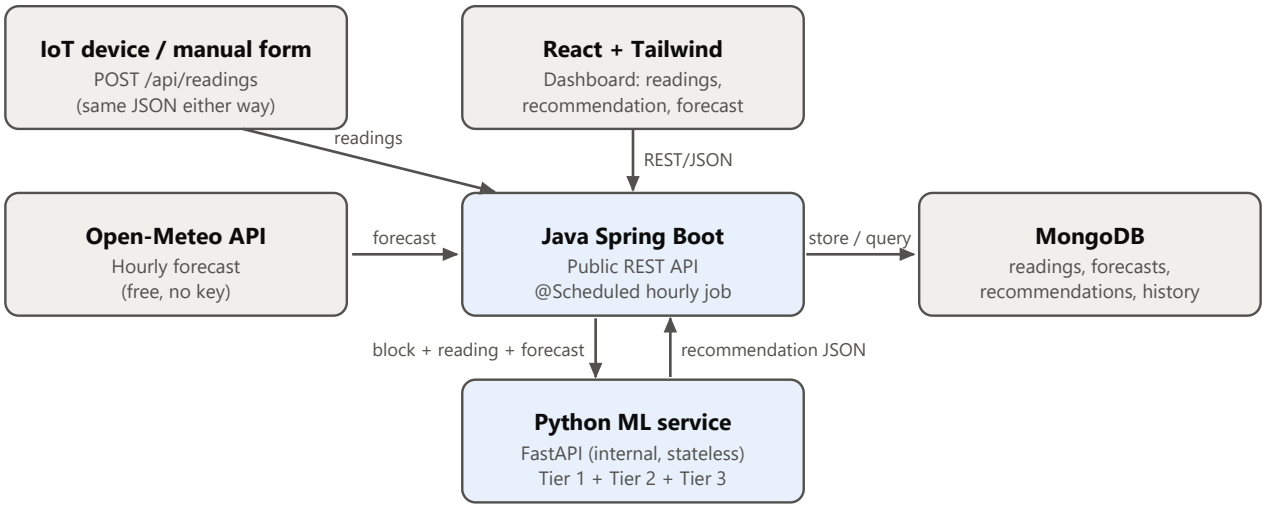


Figure 2. Software architecture. Spring Boot is the only public API and owns MongoDB; the Python ML service is internal and stateless; React talks only to Spring Boot.

Table 4. Technology stack.

Layer	Technology	Responsibility
Frontend	React.js + Tailwind CSS	Dashboard: latest readings, recommendation and reason, forecast, history
Backend	Java + Spring Boot	Public REST API; receives readings; fetches the Open-Meteo forecast; hourly `@Scheduled` re-recommendation
Database	MongoDB	Readings, forecasts, recommendations and history
ML service	Python, FastAPI, XGBoost, scikit-learn, pymoo, Optuna, SHAP	Tier 1–3 logic; receives block + reading + forecast, returns the recommendation

4. Data

4.1 Why a hybrid dataset

Training the surrogates needs block-level records of soil state, interventions, weather after each intervention and the yield that followed. No such public data exists for Sri Lanka. A fully synthetic dataset (version 1) was built first, but it had a basic weakness: every input, including the weather, came from the same equations the models then learned. The current **hybrid** dataset keeps the simulation only where no real data exists.

Table 5. What is real and what is simulated in the hybrid dataset.

Part of the data	Real or simulated	Source
Daily weather (rain, temperature, humidity, wind, radiation, ET_0)	**Real**	ERA5 via Open-Meteo, 5 locations, 2021–2025
Rain in the 4/12/24/48/72 h after each field action	**Real**	Hourly ERA5 rainfall
Rain forecast available at the time of each action	**Real** from 20 Jan 2024; earlier sampled from the real error distribution	Open-Meteo Previous Runs
Block soil: pH, total N, organic carbon, texture, CEC	**Real**	SoilGrids at each block's coordinates
Field capacity and wilting point	Derived from real texture	Saxton & Rawls (2006)
Block temperature	Real, adjusted for block elevation	−0.65 °C per 100 m
Fertilizer / irrigation / fungicide / dolomite decisions	Simulated	Historical fixed-calendar practice
Growth, yield, runoff loss, sensor noise	Simulated	Process model (Section 4.4)

4.2 Study locations

Five representative tea-growing locations cover all three elevation bands, with 10 blocks each (50 blocks). The estate names are fictional; the locations and their weather are real.

Table 6. Study locations.

ID	Location	Zone	Grid elevation	Mean annual rain (2021–25)
E01	Nuwara Eliya	Up-country (high grown)	1,868 m	2,679 mm
E02	Hatton / Dimbula	Up-country (high grown)	1,288 m	4,107 mm
E03	Kandy	Mid-country (medium grown)	632 m	1,719 mm
E04	Ratnapura	Low-country (low grown)	27 m	4,023 mm
E05	Deniyaya / Matara	Low-country (low grown)	424 m	2,660 mm

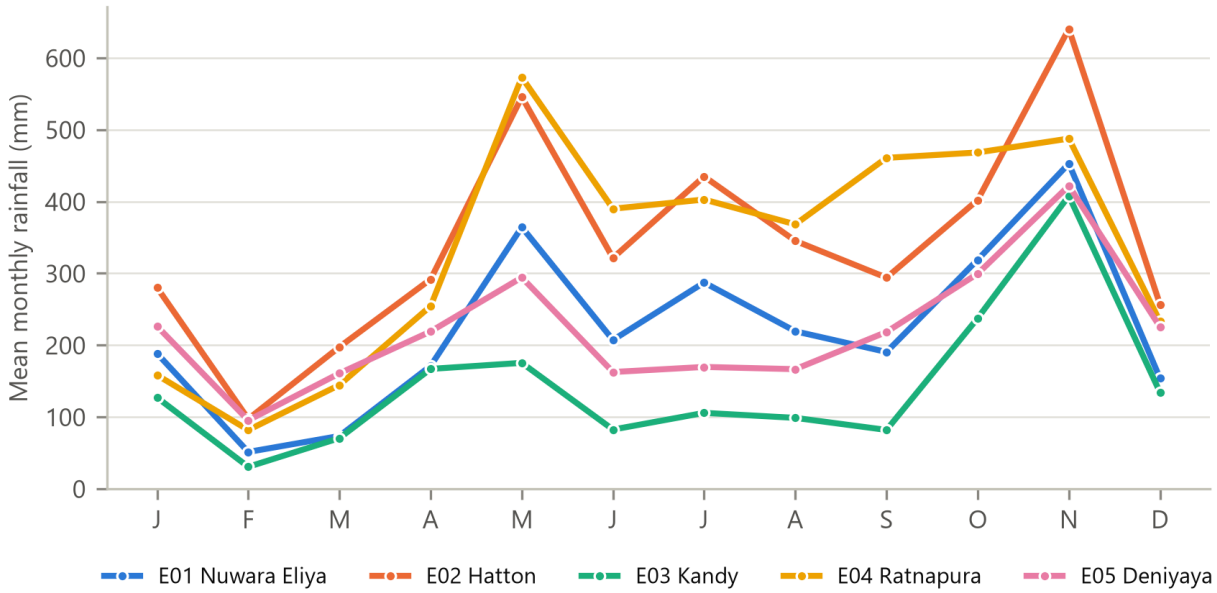


Figure 3. Mean monthly rainfall at the five locations (ERA5, 2021–2025). The two monsoon peaks (May and October–November) and the dry February are clear; Kandy (E03) is the driest location.

Block properties span realistic ranges: elevation 119–2,087 m, slope 8–45 % (median 31 %), SoilGrids pH 4.9–5.5, field capacity 38–40 % and wilting point 22–28 % (clay-rich soils). 19 of the 50 blocks have irrigation available.

4.3 Real data pack

Table 7. Real data pack (SmartTea_LK_C04_real_data).

File	Rows	Content
01 weather_hourly_era5	219,120	Hourly weather, soil moisture and soil temperature per estate
02 weather_daily_era5	9,130	Daily aggregates of file 01
03 forecast_vs_actual_hourly	85,475	Rain forecast 1–4 days ahead vs actual (from 19 Jan 2024)
04 forecast_vs_actual_daily	3,560	Daily totals of file 03
05 nasa_power_daily	9,130	Independent cross-check
06 soilgrids_block_soil	50	Soil properties at 0–5, 5–15, 15–30 cm with 5–95 % range
07 sl_tea_production_monthly	300	National production by elevation, from 47 Forbes & Walker reports
08 monthly_weather_vs_production	180	Monthly weather per zone joined with national production

4.4 Process model (simulated part)

Each block is simulated day by day. The main components are:

- **Soil water:** a bucket model with a 400 mm root zone; slope-dependent infiltration; evapotranspiration (FAO ET_0 × 0.9 × stress factor); drainage above field capacity.
- **Daily growth:** scale × age × cultivar × EWMA($f_{temp} \cdot f_{water} \cdot f_{light} \cdot f_N \cdot f_K \cdot f_P \cdot f_{pH}$) × f_{prune} × (1 − blight loss) × noise. Temperature optimum 18–26 °C; water stress below WP + 0.45·(FC − WP); nitrogen response $0.5 + 0.5 \cdot (1 - e^{-N/30})$; pH optimum 4.5–5.5; no harvest for 60 days after pruning.

- **Calibration:** mean annual green leaf ≈ 7,000 (up), 6,000 (mid) and 9,500 (low) kg/ha. The generated data matches: 7,000 / 5,999 / 9,501 kg/ha.
- **Fertilizer loss:** $0.04 + 0.62 \cdot (1 - e^{-\text{rain}_{\text{eff}}/40}) \cdot (0.7 + \text{slope}/40)$, where $\text{rain}_{\text{eff}} = r_{12} + 0.5 \cdot (r_{48} - r_{12}) + 0.2 \cdot (r_{72} - r_{48})$, plus extra loss on saturated or dry soil and in midday heat; clipped to 2–90 %.
- **Nutrients and pH:** the N pool gains fertilizer × (1 − loss) and mineralisation, and loses uptake, rain-driven leaching and immobilisation. N fertilizer acidifies the soil; dolomite raises pH over 90 days.
- **Blister blight** (above 600 m): risk from 5-day wetness × low sunshine; fungicide protects for 10 days unless more than 8 mm of rain falls within 4 h of spraying.
- **Historical management:** a fixed fertilizer calendar (up/mid: Mar, Jun, Oct; low: Feb, May, Aug, Nov) with jitter; irrigation when soil is dry; dolomite below pH 4.45; fungicide when blight risk is high.

4.5 Generated datasets

Table 8. Hybrid dataset files (SmartTea_LK_C04_hybrid_data/data).

File	Rows	Content
01_block_master	50	Static block features + SoilGrids soil, FC/WP, temperature offset
02_weather_daily	9,130	Real ERA5 weather; sunshine derived from real radiation
03_sensor_readings_daily	91,300	Same fields as the manual input form; noise and ~1.5 % missing values
04_yield_plucking_records	9,053	Green leaf kg/ha per plucking round
05_intervention_records	3,233	Fertilizer 850, fungicide 1,771, irrigation 611, dolomite 1; actual and forecast rain after each
06_resource_usage_monthly	1,200	Allocated vs used per estate and resource
07_operational_constraints	25	Optimizer limits and scheduler rules
08_daily_forecasts_issued	9,130	Day 0–3 rain forecast issued each morning (3,545 real days)
ml_yield_training	9,053	Yield model table; target green_leaf_kg_per_ha_per_day
ml_runoff_training	850	Runoff model table; target runoff_loss_est_pct

4.6 Data quality checks

All files were checked on 8 October 2026 after the repository was reorganised:

- Every row count matches the documented size; dates run 2021-01-01 to 2025-12-31 with no invalid dates.
- No duplicate rows and no duplicate (block, date) or (estate, date) keys; every block_id exists in the block master.
- Sensor values are in realistic ranges: pH 4.3–5.7, soil moisture 21–50 %, soil temperature 10–27 °C.
- Missing values are explained: ~1.5 % deliberate sensor gaps; intervention fields that only apply to one action type (e.g. irrigation depth on irrigation rows only); 10 SoilGrids blocks on masked pixels filled with the estate mean.
- All 47 Forbes & Walker PDFs are valid; national monthly totals add up across elevations to within 2 kg.

5. Methodology

5.1 Model training and validation

- **No random splits.** Random splits leak information between neighbouring days and between rounds of the same block. Two honest splits are used: a **time split** (train 2021–2024, test 2025) and a **block-grouped split** (GroupKFold by block, 5 folds), which measures performance on blocks the model has never seen.
- **Leakage control.** In the yield table, green_leaf_kg_per_ha is dropped (it is the target times the round length) and aspect is dropped (it has no effect in the generator but was being memorised as a block identifier).
- **Live-realistic runoff model.** Trained on forecast rain (fc_rain_*), not on rain that actually fell (rain_next_*). Both versions are reported so the cost of forecast error is visible.
- **Tuning.** TimeSeriesSplit with Optuna (50–100 trials); learning curves to show the data size is sufficient.
- **Monotonic constraints** where the science is clear: more rain after application must never reduce predicted runoff.
- **Explanation.** SHAP values for the surrogate models; models saved with joblib for the ML service.
- **Metrics.** R^2, MAE and RMSE, reported for both splits and both rain sources.

5.2 Evaluation plan

- **Scheduler vs fixed calendar** on the hybrid data: fertilizer and water used, runoff avoided, yield.
- **Forecast-error impact:** compare decisions made with forecasts against decisions made with perfect knowledge.
- **Sensitivity analysis** on the key placeholder parameters (fertilizer rates, loss coefficients, thresholds).
- **External validation** against real estate fertilizer and yield records, if TRI or a plantation company can provide them.

6. Preliminary results

The results below come from the baseline validation (`validate_hybrid.py`). They use default hyper-parameters; tuned models will follow in the next stage.

6.1 Surrogate models

Table 9. Baseline results: R^2 (MAE in brackets; kg/ha/day for yield, percentage points for runoff).

Test	Ridge	Random Forest	XGBoost
Yield: time split (test year 2025)	0.644 (3.29)	0.890 (1.84)	**0.939 (1.39)**
Yield: unseen blocks (5-fold GroupKFold)	–	–	**0.661**
Runoff: actual rain after (hindsight)	0.738 (6.40)	0.927 (4.26)	**0.937 (4.10)**
Runoff: forecast rain only (live-realistic)	0.473 (11.05)	0.614 (9.24)	**0.617 (9.10)**

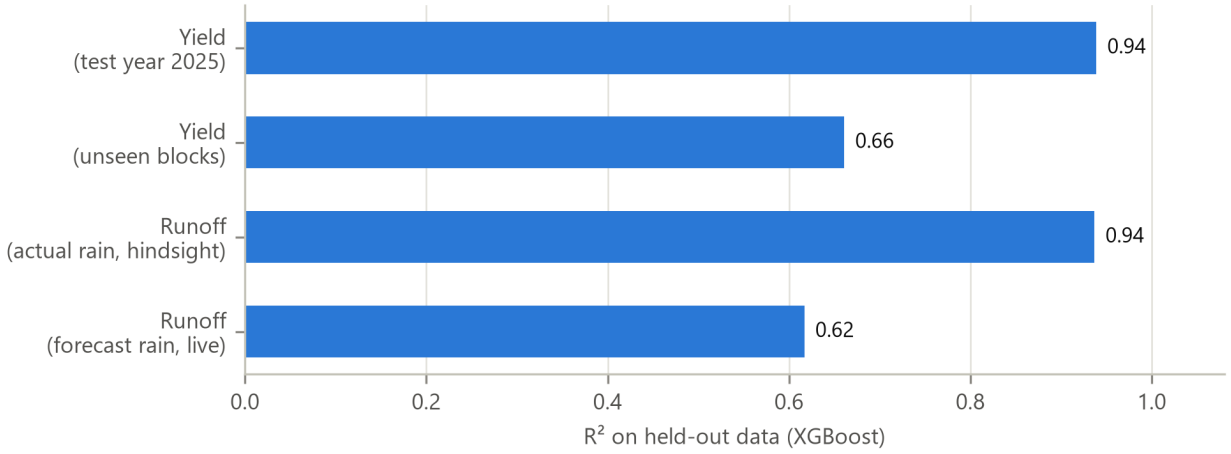


Figure 4. XGBoost R^2 under each test. The honest numbers are the second and fourth bars: performance on unseen blocks and with forecast rain only.

XGBoost is the best model in every test, as expected for tabular data of this size. Two drops matter more than the headline numbers:

- **Unseen blocks (0.94 → 0.66).** With real soil and weather, blocks differ more than in the synthetic version (where the same test gave 0.80). The model generalises to new time periods better than to new places, so block-specific readings are important for live use.
- **Forecast rain (0.94 → 0.62).** The runoff model is only as good as the rain information it receives. This is the main finding so far and is examined below.

Re-running the validation with the current package versions (XGBoost 3.4.1) gives the same Ridge and Random Forest scores and XGBoost scores within ±0.02 (e.g. 0.675 for unseen blocks), because the script does not fix XGBoost's random seed.

6.2 Real forecast skill

Comparing archived Open-Meteo forecasts with what actually fell at the five locations (2024–2025):

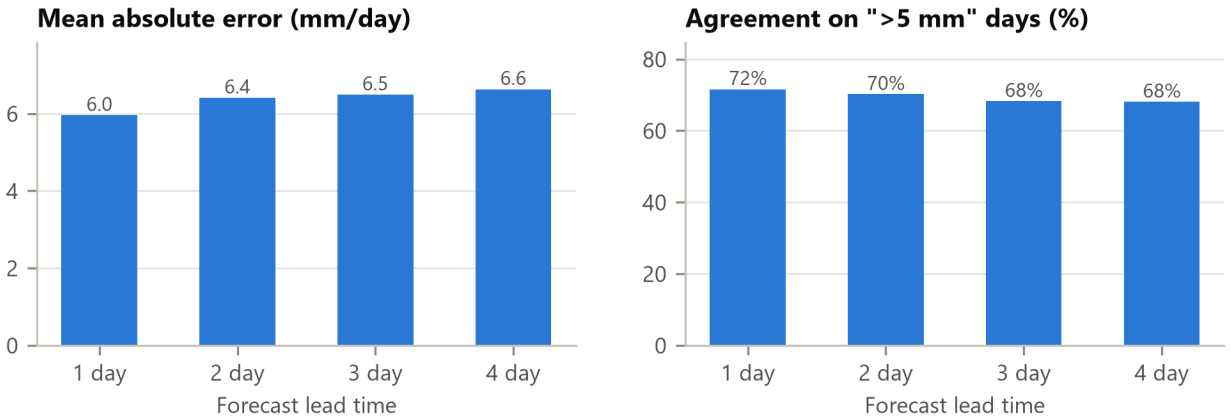


Figure 5. Daily rain forecast skill by lead time. Error grows and agreement falls after day 1, exactly inside the 72–96 h window the scheduler must use.

Table 10. Forecast skill for daily rainfall at the five estates, 2024–2025.

Lead time	MAE (mm/day)	Bias (mm/day)	Agreement on “>5 mm” days	Correlation
1 day	6.0	−3.6	72 %	0.69
2 days	6.4	−3.9	70 %	0.59
3 days	6.5	−4.3	68 %	0.50
4 days	6.6	−4.0	68 %	0.55

Key finding. Forecasts consistently **under-predict** rain (bias about −4 mm/day), and roughly one day in three is misclassified as wet or dry even one day ahead. A scheduler that trusts the forecast will sometimes apply fertilizer just before rain it was not warned about. This explains most of the drop in runoff-model accuracy and argues for safety margins, rain-probability thresholds and a local forecast-correction model.

6.3 Runoff and rain after application

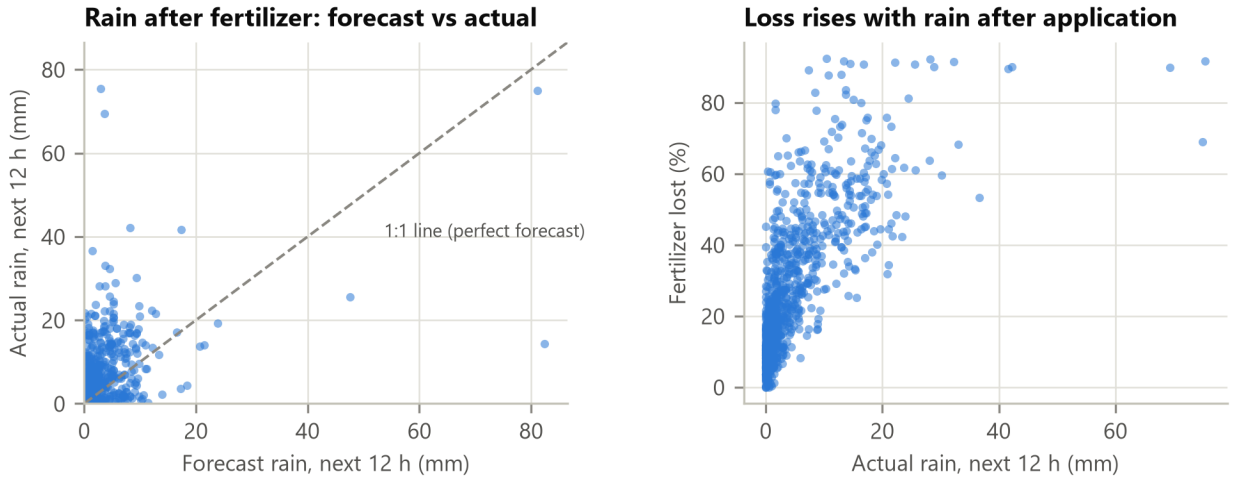


Figure 6. Left: forecast vs actual rain in the 12 h after each of the 850 fertilizer applications (correlation 0.46); many points sit above the 1:1 line, where more rain fell than was forecast. Right: the simulated fertilizer loss rises steeply with the rain that follows.

Across the 850 fertilizer applications under the historical fixed calendar, the median loss is 24 % and one application in ten loses more than 59 %. This is the waste the Action-Window scheduler aims to reduce. Of these 850 events, 340 (40 %) use real archived forecasts; the rest use forecasts sampled from the real error distribution.

6.4 Realism checks

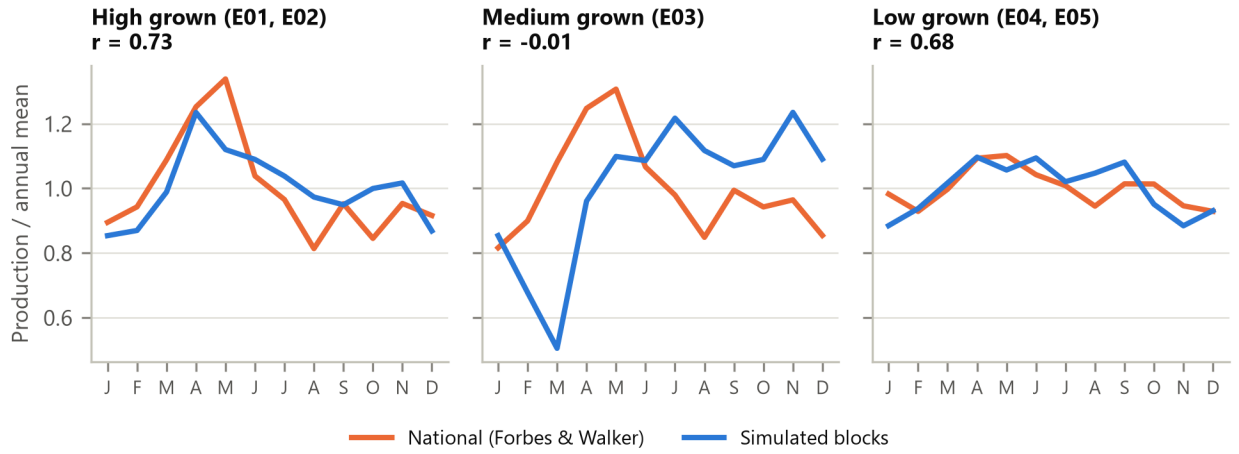


Figure 7. Seasonal shape of simulated block yield vs national production for each elevation band (each series divided by its own annual mean). High and low grown follow the national pattern; medium grown does not.

- **Soil moisture:** the simulated soil moisture correlates with ERA5-Land soil moisture at r = 0.85–0.91 at all five locations.
- **Seasonality:** high-grown r = 0.73 and low-grown r = 0.68 with national production. **Medium-grown r ≈ 0**: the simulated Kandy blocks dip in February–March (Kandy's dry months) while national medium-grown output peaks in March–May. Changing root depth (600–1,000 mm) and wilting point (×0.65–0.85) did not fix it. One location may not represent all medium-grown districts, or the growth model's drought response may be too strong.
- **Fertilizer response:** median yield change after fertilizer is +10.7 %, close to the +9.2 % example in the original framework.

6.5 Synthetic (v1) vs hybrid

Table 11. Effect of replacing the simulated environment with real data.

Test	Synthetic v1	Hybrid (current)
Yield, XGBoost, test year 2025	R^2 0.93	R^2 0.94
Yield, XGBoost, unseen blocks	R^2 0.80	R^2 0.66
Runoff, actual rain (hindsight)	R^2 0.93	R^2 0.94
Runoff, forecast rain only	not possible	R^2 0.62
Soil moisture vs ERA5-Land	–	r 0.85–0.91

Lower scores in the hybrid version are an improvement in honesty, not a regression: the real environment makes the task harder in the same way the live system will find it hard.

7. Discussion and limitations

These limitations are stated openly and will be carried into the thesis.

- **The data is a hybrid.** Weather, rain timing, forecasts and soil are real; management decisions, crop response, runoff amounts and sensor noise are simulated. The models therefore learn the coded process rules on top of a real environment. High R^2 reflects those rules, not proven agronomy. Validation on real estate records is the main open step.
- **Agronomic parameters are placeholders.** Fertilizer rates (~240 kg/ha mixture per application, max 400), loss coefficients, the blight model, dolomite and fungicide amounts and the 7-day pre-harvest interval must be checked against TRI guidance.
- **The runoff target cannot be measured directly** in the field. The "absorption / efficiency index" still needs a formal definition and a way to measure it.
- **Small tables.** The runoff model has 850 rows, and there is only **one dolomite event**, because real SoilGrids pH (~5.2) is already in the target band. pH correction will need a rule-based component or a regenerated dataset; it cannot be learned from this data.
- **Medium-grown seasonality mismatch** (Section 6.4). Treat mid-country results with caution.
- **2022 is unusual.** National production was hit by the fertilizer import ban and the economic crisis; this is not weather-driven.
- **Gridded data.** ERA5 is a ~9–25 km reanalysis: mountain rainfall and temperature are smoothed, and Deniyaya rainfall looks low. SoilGrids nitrogen is total N, not plant-available N (approximated as 1.2 %); pH is pH in water.
- **Forecasts before 2024 are simulated** from the real error distribution (60 % of fertilizer events), so the forecast-based score mixes real and simulated error.
- **Tree models cannot extrapolate.** The optimizer must stay inside the training range, which limits recommendations to practices similar to those in the data.

8. Work plan and open questions

8.1 Roadmap

Table 12. Development roadmap.

#	Step	Main tasks	Status
1	Repository and environment	Git, Python environment, folder structure, tidy data paths	**Done**
–	Datasets	Real data pack, hybrid generator, validation, quality checks	**Done**
2	Model training	XGBoost yield and runoff models; Ridge/RF/LightGBM/MLP baselines; Optuna; learning curves; SHAP; monotonic constraints; joblib	Next
3	Optimizer	NSGA-II with surrogate models, constraint table and training-range bounds; Pareto selection rule	Planned
4	Action-Window scheduler	Open-Meteo forecast per estate; hourly scoring with the runoff model; rule filters	Planned
5	Backend	Python ML service (FastAPI); Spring Boot API with MongoDB; hourly @Scheduled job	Planned
6	Dashboard	React + Tailwind: readings, recommendation and reason, forecast, history	Planned
7	Evaluation	Scheduler vs fixed calendar; forecast-error impact; sensitivity analysis	Planned

8.2 Open questions

- Obtain real estate fertilizer and yield records (TRI or a plantation company) for final validation.
- Check fertilizer rates, N limits, loss coefficients, fungicide pre-harvest interval and blight parameters against TRI guidance.
- Resolve or explain the medium-grown seasonality gap (more mid-country locations, or a revised drought response).
- Decide the rule for choosing one plan from the NSGA-II Pareto front.
- Decide whether timing is optimised jointly in NSGA-II or by Tier 3 re-scoring.
- Define the absorption / efficiency index formally.
- Optional: train a local weather-correction model on the forecast-vs-actual archive to remove the dry bias.

9. Conclusion

This component has defined a complete, purely agronomic decision pipeline for tea-block interventions, built a hybrid dataset that grounds the simulation in real Sri Lankan weather, forecasts and soil, and established honest baselines. XGBoost surrogates learn the yield and runoff behaviour well within known blocks and with hindsight, but the scores fall to R^2 0.66 on unseen blocks and 0.62 with forecast rain. The second drop is the clearest message of the work so far: the benefit of smart timing is bounded by forecast quality, and a practical system must treat the forecast as uncertain. The next stages tune the models, build the NSGA-II optimizer and Action-Window scheduler, deliver them through the Spring Boot and React application, and measure the gain over fixed-calendar practice.

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Uva Medium rainfall–yield study (2003–2012), the Explainable-AI Ceylon tea yield study (ScienceDirect, 2025) and the NTHRI hybrid tea-yield study (MDPI Plants, 2022) were consulted as related work; full citations are to be added.

Appendix A. Key fields

Table 13. Key fields used by the models.

Field	Table	Meaning
block_id	all	Block key, e.g. E03-B07 (estate E03, block 7)
soil_moisture_pct, soil_ph, soil_n/p/k_ppm, soil_temp_c, light_lux, sunshine_hours	03 sensor readings	Daily readings with the same fields as the manual input form
field_capacity_pct, wilting_point_pct	01 block master	Soil water limits from Saxton–Rawls; all soil-moisture rules are relative to these
sg_*	01 block master	SoilGrids properties at the block coordinates
rain_next_4h…72h_mm	05 interventions	Rain that actually fell after the action (hindsight)
fc_rain_next_12h…72h_mm	05 interventions	Rain forecast at the time of the action (live-realistic)
forecast_source	05 interventions	Real archived forecast, or sampled from the real error distribution
runoff_loss_est_pct	05 / ml_runoff	Fertilizer lost (%); runoff model target
absorption_efficiency_pct	05 / ml_runoff	Alternative efficiency target
green_leaf_kg_per_ha_per_day	04 / ml_yield	Yield model target
yield_before / after_kg_ha_day, yield_delta_pct	05 interventions	Yield response around each action

Appendix B. Reproducing the data and validation

```
cd SmartTea_LK_C04_hybrid_data
python generate_smarttea_data.py --out ./data --real-data "../SmartTea_LK_C04_real_data"
python generate_smarttea_data.py --out ./data_synthetic        # fully synthetic mode
python validate_hybrid.py data "../SmartTea_LK_C04_real_data"

Options: --seed 42  --blocks-per-estate 10  --root-zone-mm 400  --wp-factor 1.0  --missing-rate 0.015
```

Generation takes about one minute. Environment: Python 3.14, numpy, pandas, scikit-learn, XGBoost 3.4, pymoo 0.6, Optuna 5.0, SHAP 0.52, FastAPI.